Description of Practical Assignment “Study Planner”

Team of developers

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The framework, language and environment

Newest version of PHP and Laravel (PHP 8.5 and Laravel 13). The Docker and DDEV environment will be used. The code of the project will be published and saved on GitHub.

Main idea/functionality

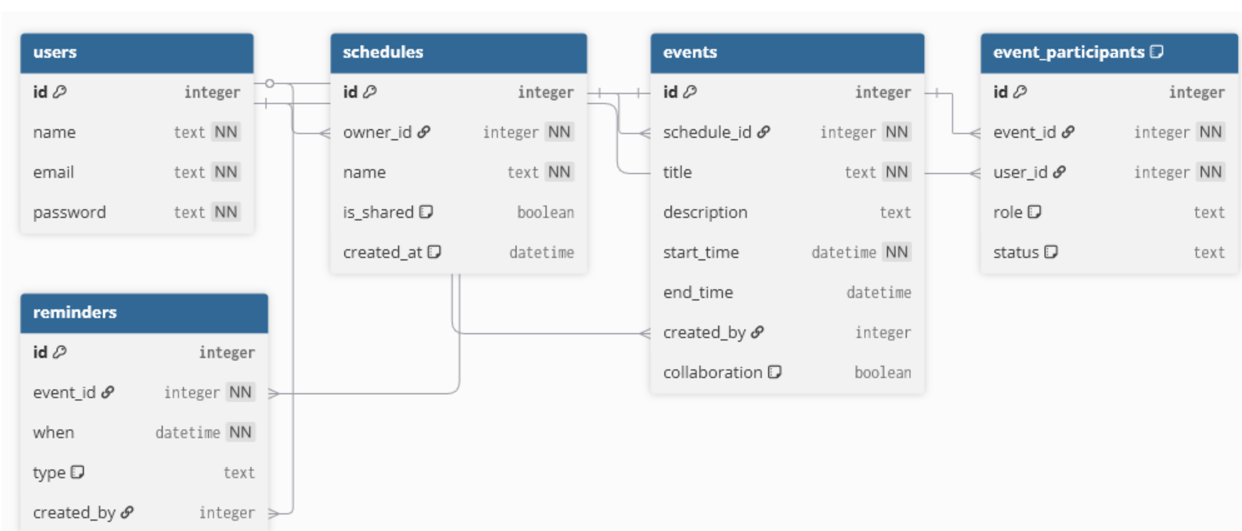
A study planner system will be developed.

It will contain such functionalities as being able to add lectures/lessons to your weekly plan with the time, homework etc. that relates to it. Schedule other events. Make reminders for schoolwork and homework. It is also planned to make collaborative planners/events between multiple people, as well as sending notifications. More functionalities will probably be made available in the making of the project.

Data Registry

The most important concepts in the system: users, schedules, events, event participants and reminders for the events.

Each user has one or multiple schedules, so they could make separate schedules. In each schedule it is possible to make events, it is possible to make reminders for the events. Each user can make a shared schedule, events in the schedule can be modified by a particular user depending on their role, which could be owner and viewer.



Relationships: schedules.owner_id > users.id, events.schedule_id > schedules.id, events.created_by > users.id, event_participants.event_id > events.id, event_participants.user_id > users.id, reminders.event_id > events.id, reminders.created_by > users.id

MVC

The system's UI will be implemented following an MVC paradigm. The system will be distributed into the following components:

Models:

- User
- Schedule
- Event
- Event participant
- Reminder

Views:

- Dashboard (can see upcoming events and reminders).
- Calendar view (can see all events in a schedule).
- Create event or schedule form view.
- Collaboration view (invite users or get invitations).
- Reminder (modal box or view).

Controllers:

- AuthenticationController (methods for creating users and user authentication).
- ScheduleController (methods for creating schedules, controlling access to them, retrieving and posting relevant info).
- EventController (methods for creating events, retrieving and posting info for calendar).
- CollaborationController (methods for adding others to events and controlling roles)
- ReminderController (methods for creating reminders and getting upcoming reminders).

User roles

The system will only support roles with sign-in, i.e. user account, it will also be possible to create schedules offline without an account, but it won't be possible to collaborate with others.

Registered users will be able to use all features, such as creating schedules and events, setting reminders, collaboration. Users will not have access to schedules and events if not invited by the creator of the respective cases or if the user has only created local schedules.

User authentication

For the user authentication it will be possible to register using an email address or *Google* authentication. If the user so pleases, it will be possible to create items in the system strictly locally.

System interface

The image below shows the planned view of the calendar. It would be a simple calendar view, a tab for each schedule. Each day could contain mutliple events, in the calendar view you could only see the name of the event and time (if a reminder is set). When clicking the “View” button the user could see the description of the event, collaboration, etc.

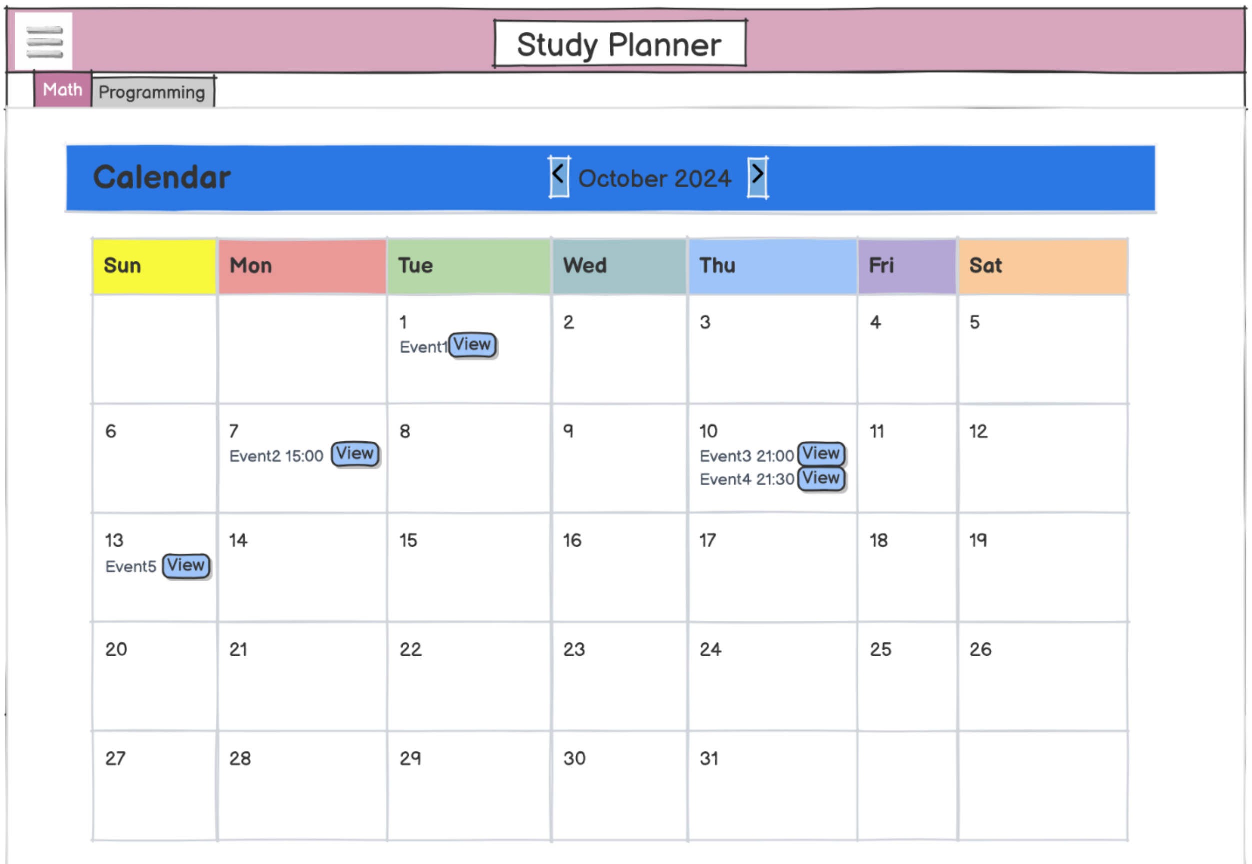
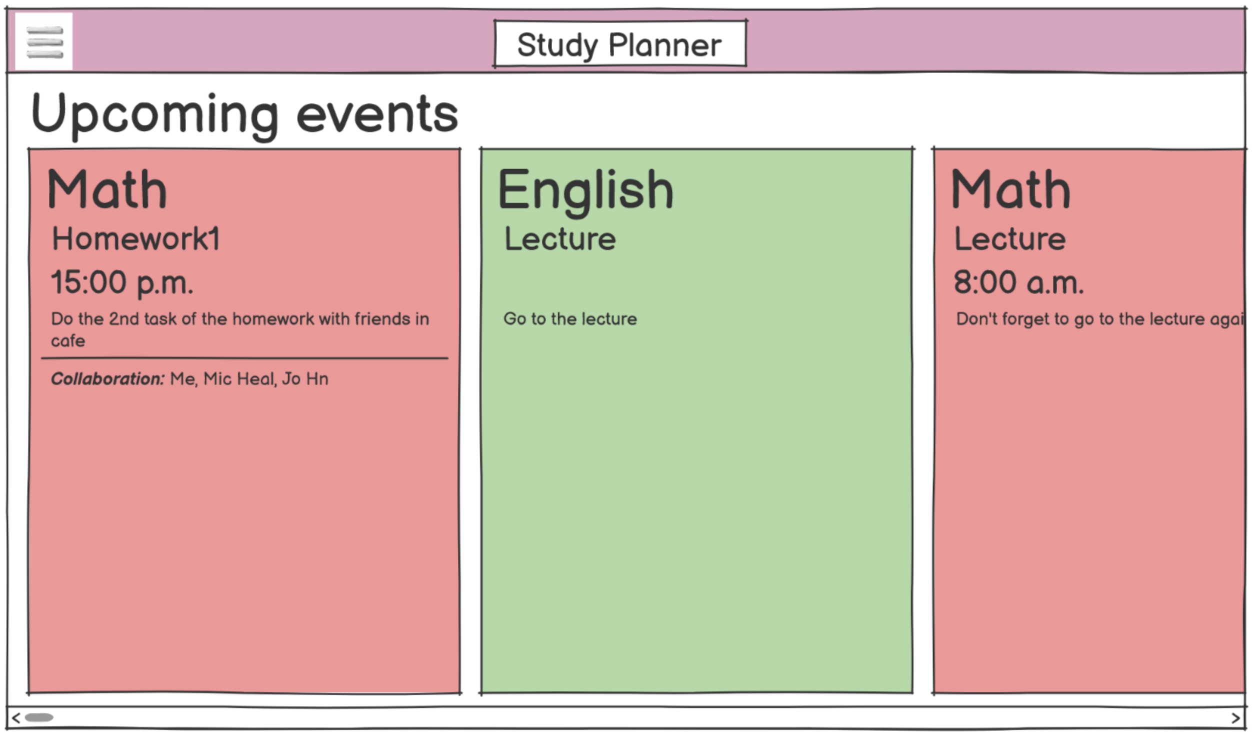


Image below shows the planned view of the dashboard. It would show cards for any schedule with superficial descriptions of the event. At the top of the card – from which schedule it is. Title below that is what event it is. Then reminder, description, etc.


Study Planner
Upcoming events
Math
Homework1
15:00 p.m.
Do the 2nd task of the homework with friends in cafe
Collaboration: Me, Mic Heal, Jo Hn
English
Lecture
Go to the lecture
Math
Lecture
8:00 a.m.
Don't forget to go to the lecture agai